

Wildfire Burned Area Prediction

A Machine Learning Approach to Mitigating Climate and Environmental Risks

ACE6313 Machine Learning Concepts and Technologies (T2610)

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Alignment with UN SDG 13: Climate Action

SDG 13 CLIMATE ACTION

Building tools for resilience, early warning, and carbon emissions reduction through forest preservation.

SDG Target 13.1 (Resilience & Adaptability): Providing prediction tools to enhance disaster preparedness and emergency response times.

SDG Target 13.3 (Early Warning & Awareness): Utilizing geographic and seasonal patterns to formulate regional and industrial warnings.

Climate Mitigation: Successful early containment prevents catastrophic carbon emissions from burning forests, protecting critical carbon sinks.

Problem Statement & Objectives

The Problem

Wildfire spread is highly chaotic, governed by weather, fuel, topography, and administrative response times.

Forecasting final fire size (EstTotalAcres) at the moment of ignition is a highly non-linear, right-skewed regression task.

Key Objectives

- 1. Preprocessing Pipeline:** Develop a clean data preprocessing pipeline (cleaning, feature extraction).
 - 2. Leakage Prevention:** Enforce strict feature selection to prevent target leakage.
 - 3. Model Comparison:** Train and compare 6 distinct regression models (Linear Regression, kNN, Decision Trees, Random Forest, Gradient Boosting, MLP).
 - 4. Dimensional Reduction:** Test PCA's efficacy for dimensional reduction on tree and neural models.
 - 5. Optimization:** Optimize the top-performing model through hyperparameter tuning.
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Data Preprocessing & Cleaning

Dataset & Cleaning

Dataset: Oregon Department of Forestry (ODF) Fire Occurrence dataset (2000–2023, 23,490 raw records).

Data Cleaning: Handled missing records, filtered coordinates, leaving 23,202 clean samples.

Target Leakage Prevention

Dropped Size_class: Size classes (A-G) are directly calculated from final acreage, creating automatic target leakage.

Dropped Control_DateTime: Containment times are only known post-event, making them unavailable at ignition.

Removed Metadata: Serial numbers, names, and unique IDs which contain no predictive value.

Feature Engineering & Encoding

Temporal Feature Engineering

Ignition DateTime Parsing: Parsed Ign_DateTime to extract seasonal and daily indicators.

Ign_Month: Captures seasonal risk profiles, representing seasonal climate cycles.

Ign_Hour: Captures peak daily temperature periods and daily cycles.

Categorical Encoding Strategy

High-Cardinality Dilemma: Applying one-hot encoding to high-cardinality geographic columns (Twn, Rng, Subdiv) creates 23,463 sparse features, leading to overfitting.

Label Encoding Solution: Applied category coding (Label Encoding) to retain a compact feature space of 24 dense features, maintaining generalizability.

Machine Learning Algorithms

Evaluation of 6 Regressors

Parametric Baseline: Linear Regression

Instance-Based: k-Nearest Neighbors (kNN)

Non-Parametric Trees: Decision Trees, Random Forest, and Gradient Boosting

Neural Network: Multi-Layer Perceptron (MLP)

Feature Scaling

Standardization: Standardized all continuous and encoded variables using StandardScaler.

Reasoning: Ensures equal feature weighting for distance (kNN) and weight-based gradient updates (MLP) to avoid scaling bias.

Feature Reduction via PCA

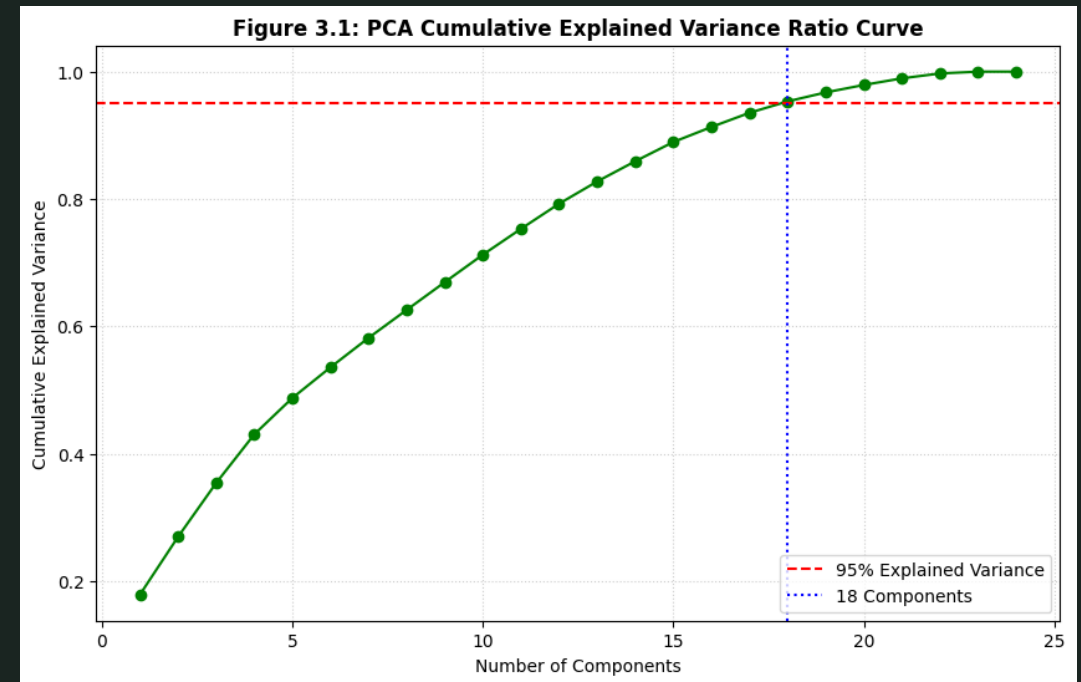
PCA Setup & Key Findings

PCA Configuration: Applied Principal Component Analysis targeting 95% explained variance.

Compression: Reduced the feature space from 24 features to 18 orthogonal components.

Severe Tree Degradation: Random Forest R^2 fell from +0.1220 (Standard) to -0.0841 (PCA).

Why? PCA creates linear combinations of features, which destroys the axis-aligned splits and local boundaries that decision trees rely on.



Model Performance Comparison

Model / Algorithm	Standard R ²	Standard RMSE (Acres)	PCA R ²
Random Forest (Top Model)	0.1220	6951.77	-0.0841
Linear Regression	-0.0088	7451.70	-0.0092
MLP (Neural Net)	-0.0168	7481.42	-0.0125
kNN	-0.0284	7523.84	-0.0342
Gradient Boosting	-0.0419	7572.92	-0.0144
Decision Tree	-1.1923	10985.28	-0.2132

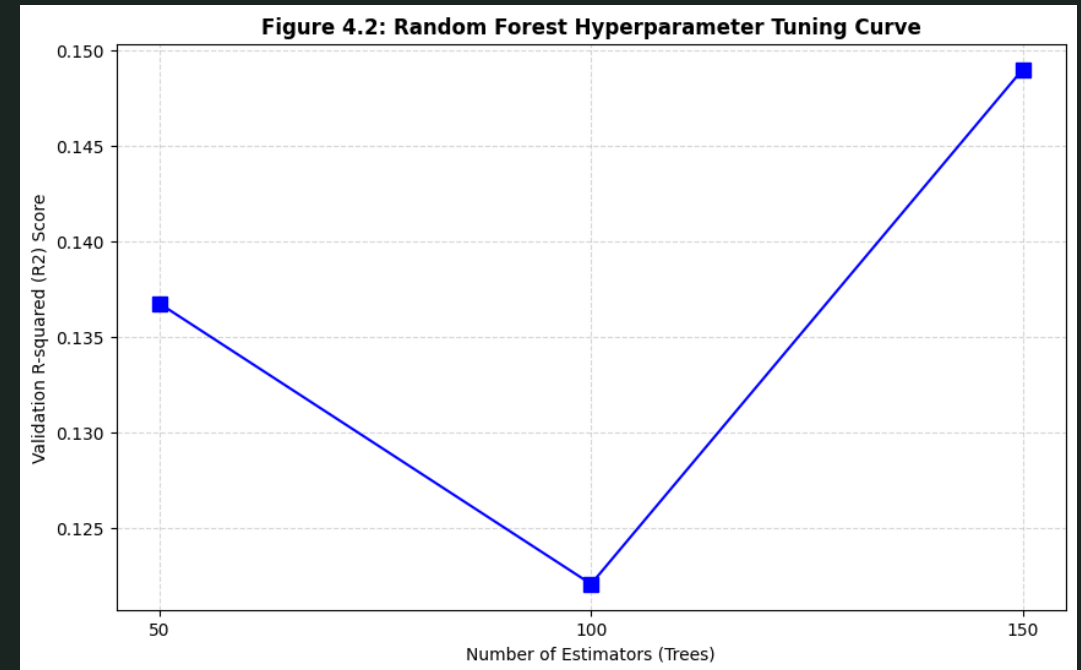
Random Forest Hyperparameter Tuning

Optimizing Ensemble Size

Ensemble Size ($n_estimators$): Evaluated the impact of ensemble size on prediction performance:

- **50 trees:** $R^2 = 0.1367$
- **100 trees (Baseline):** $R^2 = 0.1220$
- **150 trees (Best):** $R^2 = 0.1490$ (RMSE = 6844.21 acres)

Performance Summary: Optimization yielded a +2.7% baseline improvement in R^2 and reduced the prediction error (RMSE).



Selected Final Model & Configuration

Chosen Model Configuration

Model Selection: Random Forest Regressor

Optimized Hyperparameter: `n_estimators` = 150 (150 decision trees)

random_state: 42 (for absolute reproducibility)

criterion: 'squared_error' (mean squared error loss)

max_depth: Unlimited (grown fully to capture complex thresholds)

Final Evaluation Results

Final R² Score: 0.1490 (+2.7% relative improvement over baseline)

Final RMSE: 6844.21 acres (reduced from 6951.77 acres)

Key Strength: Superior generalizability over non-ensemble architectures on non-linear ODF features.

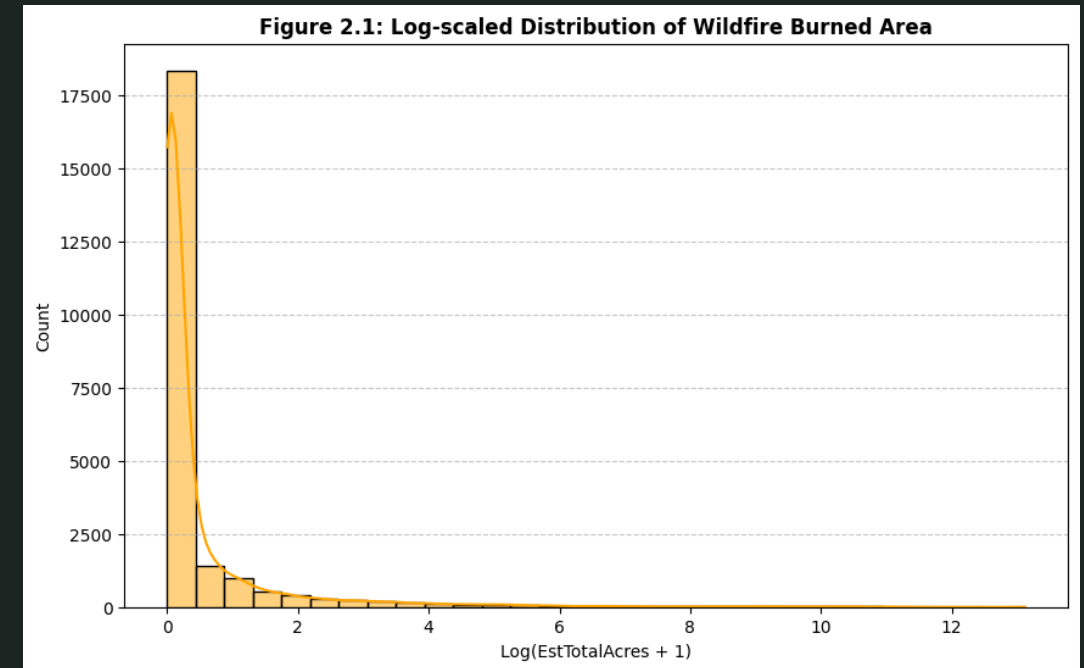
Results Discussion & Challenges

Key Challenges

Highly Skewed Target Distribution: Wildfires follow a power-law distribution. Most fires are very small (< 1 acre), but a few massive fires account for the bulk of burned acres.

MSE/RMSE Penalty: Standard regression loss functions (MSE/RMSE) quadratically penalize errors on these extreme outliers, driving high volatility.

Missing Meteorological Predictors: Administrative data is static. Predicting fire spread requires dynamic parameters like wind speed, fuel dryness, temperature, and relative humidity.



Conclusion & Future Work

Key Achievements

Leakage-Free Pipeline: Developed a leakage-free ML pipeline; identified Random Forest as the superior regressor; analyzed PCA's limitations on tree ensembling.

Proposed Future Work

Log Transformation: Train models on $\log(\text{Acres} + 1)$ to compress skewed outliers.

Weather Data Integration: Merge historical fire coordinates with meteorological data (NOAA, ERA5).

Classification Reframing: Predict discrete fire size categories (A through G) for actionable resource staging.

Team Contributions & Reflections

Lwee Tong Jun (Data Preprocessing)

Contributions: Target leakage discovery, datetime engineering, categorical coding design.

Key Learning: Practical leakage resolution and categorical representation trade-offs in real-world GIS databases.

Peng Ze Wei (ML Models)

Contributions: Implementation of 6 regressors, scaling, PCA design, RF tuning.

Key Learning: Understanding PCA's destructive effect on non-linear decision boundaries and interpreting regression metrics in skewed datasets.

Thank You!

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Lwee Tong Jun & Peng Ze Wei
Questions & Answers